

- plinaste
- trdne (zlitine)
- tekoče

topilo: kom. 1
topilnica: kom. 2

Sestava raztopin

- molarna koncentracija $c = \frac{n_2}{V}$ ← množina toplicev v raztopini
- molalna koncentracija $b = \frac{n_2}{m_1}$ ← masa topila
- masna koncentracija $f = \frac{m_2}{V}$ ← masa toplicev
- molski delež $x_2 = \frac{n_2}{n} = \frac{n_2}{n_1 + n_2}$
- masni delež $w = \frac{m_2}{m} = \frac{m_2}{m_1 + m_2}$
- gostota $\rho = \frac{m_1 + m_2}{V}$

$$u_i = \frac{m_i}{M_i}$$

$$(dG)_{T,p} = \sum \mu_i dn_i = \mu_1 dn_1 + \mu_2 dn_2$$

$$G = \int_0^1 dG = \int_0^1 \mu_1 dn_1 + \int_0^1 \mu_2 dn_2 = \int_0^1 \mu_1 n_1 d\lambda + \int_0^1 \mu_2 n_2 d\lambda = n_1 \mu_1 + n_2 \mu_2 \quad / \cdot d$$

$$dG = n_1 d\mu_1 + dn_1 \mu_1 + n_2 d\mu_2 + dn_2 \mu_2$$

$$n_1 d\mu_1 + n_2 d\mu_2 = 0$$

Gibbs-Duhéné

Parcialne molske količine

$$\bar{X}_i = \left(\frac{\partial X}{\partial n_i} \right)_{T,p,n_j}$$

$$\sum_i n_i d\bar{X}_i = 0$$

Gibbs-Duhéné

$X = G, H, U, A, S, V$

$$X = n_1 \bar{X}_1 + n_2 \bar{X}_2$$

Primer: NaCl v vodi

$$V = A + Bb + Cb^{3/2} + Db^2$$

$$A = 100,138 \text{ mL}$$

$$B = 16,625 \text{ mL} \frac{\text{kg}}{\text{mol}}$$

$$C = 1,7735 \text{ mL} \left(\frac{\text{kg}}{\text{mol}} \right)^{3/2}$$

$$D = 0,1194 \text{ mL} \left(\frac{\text{kg}}{\text{mol}} \right)^2$$

$$b = 0,5 \frac{\text{mol}}{\text{kg}}, n_1 = 1 \text{ kg}$$

$$\bar{V}_1(b) = ?$$

$$\bar{V}_2(b) = ?$$

$$\bar{V}_2 = \left(\frac{\partial V}{\partial n_2} \right)_{T,p,n_1} = \left(\frac{\partial V}{\partial b} \right)_{T,p,n_1} \left(\frac{\partial b}{\partial n_2} \right) = \frac{1}{m_1} \left(B + \frac{3}{2} C b^{1/2} + 2 D b \right) = 18,626 \frac{\text{mL}}{\text{mol}}$$

$$n_1 d\bar{V}_1 + n_2 d\bar{V}_2 = 0$$

$$d\bar{V}_1 = - \frac{n_2}{n_1} d\bar{V}_2, \quad \frac{n_2}{n_1} = b M_1$$

$$d\bar{V}_1 = - \frac{b M_1}{n_1} \left(\frac{3}{2} C b^{1/2} + 2 D \right) db$$

$$\bar{V}_1 - V_{1,n} = - \frac{n_1}{n_1} \left(\frac{1}{2} C b^{3/2} + D b^2 \right)$$

$$\bar{V}_1 = V_{1,n} - \frac{n_2}{n_1} \left(\frac{1}{2} C b^{3/2} + D b^2 \right) = 18,047 \frac{\text{mL}}{\text{mol}}$$

Primer

$$V_2 = 100 \text{ mL propan-1-ol}$$

$$V_1 = 100 \text{ mL H}_2\text{O}$$

$$M_2 = 60,09 \text{ g/mol}$$

$$M_1 = 18,02 \text{ g/mol}$$

$$\rho_2 = 0,998 \text{ g/L}$$

$$\rho_1 = 0,993 \text{ g/L}$$

$$V = ?$$

$$n(\text{H}_2\text{O}) = n_1 = \frac{m_1}{M_1} = \frac{V_1 \rho_1}{M_1} = 5,538 \text{ mol}$$

$$n_2 = 1,336 \text{ mol}$$

$$\bar{V}_1 = 18 \frac{\text{mL}}{\text{mol}}$$

$$\bar{V}_2 = 72 \frac{\text{mL}}{\text{mol}}$$

$$\rightarrow V = n_1 \bar{V}_1 + n_2 \bar{V}_2 = 196 \text{ mL}$$

$$\bar{X}_i = f(\text{koncentracija})$$

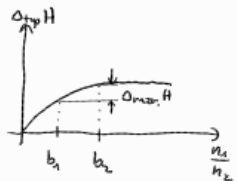
$$\Delta_{\text{mes}} H = H(\text{razt.}) - \sum_i H_i(\text{kom.}) =$$

$$= n_1 \bar{H}_1 + n_2 \bar{H}_2 - (n_1 H_{1,n} + n_2 H_{2,n}) =$$

$$= n_1 (\bar{H}_1 - H_{1,n}) + n_2 (\bar{H}_2 - H_{2,n})$$

relativna
molska
entalpija

$$\Delta_{\text{top}} H = \frac{\Delta_{\text{mes}} H}{n_2} = \frac{n_1}{n_2} (\bar{H}_1 - H_{1,n}) + (\bar{H}_2 - H_{2,n})$$



$$\Delta_{\text{top}} \bar{H} = \left(\frac{\partial \Delta_{\text{mix}} H}{\partial n_2} \right)_{n_1} \text{ diferencialna toplota topila}$$

$$\Delta_{\text{mez}} \bar{H} = \left(\frac{\partial \Delta_{\text{mix}} H}{\partial n_1} \right)_{n_2} \text{ diferencialna razredilna toplota}$$

$$d(\Delta_{\text{mix}} H) = (\bar{H}_1 - H_{1,m}) dn_1 + n_1 d(\bar{H}_1 - H_{1,m}) + (\bar{H}_2 - H_{2,m}) dn_2 + n_2 d(\bar{H}_2 - H_{2,m})$$

Gibbs-Duhem

$$\left(\frac{\partial G}{\partial n_i} \right)_{T, p, n_j \neq i} = \mu_i = \bar{G}_i$$

$$d\mu_i = d\bar{G}_i = -\bar{S}_i dT + \bar{V}_i dp$$

$$\left(\frac{\partial (\mu_i / T)}{\partial T} \right)_p = - \frac{\Delta \bar{H}_i}{T^2}$$

Idealna raztopina

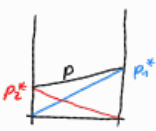
$$\Delta_{\text{mix}} H = 0$$

$$\Delta_{\text{mix}} V = 0$$

1 in 2 sta tekoča, hlapna, mešata se v vseh razmerjih

Raoult

$p_i = x_i p_i^*$ ← parni tlak čiste komp. i



$$p = x_1 p_1^* + x_2 p_2^*$$

$$x_1 + x_2 = 1$$

$$p = x_1 p_1^* + (1 - x_1) p_2^* = p_2^* + x_1 (p_1^* - p_2^*)$$

$$x_1 = \frac{n_1}{n_1 + n_2}$$

$y_1, y_2 \dots$ sestava parne faze

$$y_1 = \frac{p_1}{p} = \frac{p_1^* x_1}{p}$$

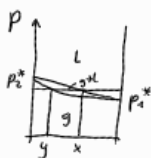
↳ molski delež komponente 1 v plinasti fazi

$$x_1 = \frac{y_1 p}{p_1^*}$$

$$p = p_2^* + \frac{y_1 p}{p_1^*} (p_1^* - p_2^*) \quad / \cdot p_1^*$$

$$p p_1^* = p_1^* p_2^* + y_1 p (p_1^* - p_2^*)$$

$$p = \frac{p_1^* p_2^*}{p_1^* - y_1 (p_1^* - p_2^*)}$$

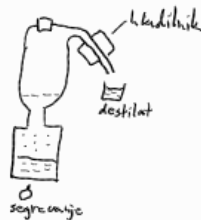


Destilacija

• vzgledno parno

• je utekočinjeno, vzpostavi se nova ravnovesje

• pojavljanje



Primer

propan-1-ol ①

$$T = 25^\circ \text{C}$$

$$p = p_2^* + x_1 (p_1^* - p_2^*) = 35,5 \text{ torr}$$

propan-2-ol ②

$$p = ?$$

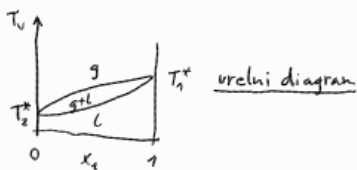
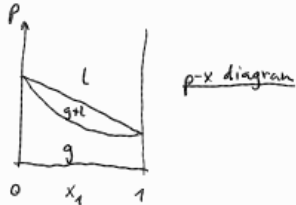
$$y_1 = \frac{p_1}{p} = \frac{x_1 p_1^*}{p} = 0,236$$

$$x_1 = 0,4$$

$$y_1 = ?$$

$$p_1^* = 20,9 \text{ torr}$$

$$p_2^* = 45,2 \text{ torr}$$



Neidealne raztopine

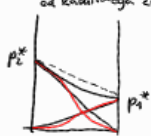
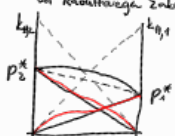
za koncentrirano komponento Raoult: $p_i = x_i p_i^*$

za razredčeno komponento Henry: $p_i = x_i k_{H,i}$

Henryjeva konstanta

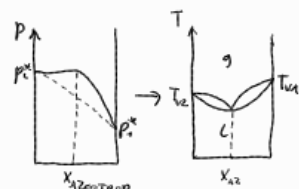
1) $k_{H,i} > p_i^*$
(pozitivno odstopanje od Raoultovega zakona)

2) $k_{H,i} < p_i^*$
(negativno odstopanje od Raoultovega zakona)



skrajni primer: maksimni tlaka je nekje vmes

⇒ 2 destilacije ne moremo ločiti in komponente



Azeotropna zmes

Odnosnost $\mu(x)$

$$\mu_i = \mu_i^* + RT \ln \frac{f_i}{p^*}$$

Raztopina:

$$\mu_i(l) = \mu_i(g)$$

Za čisto komponento

$$\mu_i^*(l) = \mu_i^* + RT \ln \frac{f_i^*}{p^*}$$

ken. pot. čiste snovi

$$\mu_i(l) - \mu_i^*(l) = RT \ln \frac{f_i}{f_i^*} \leftarrow \text{fugativnost čiste snovi}$$

$$\mu_i(l) = \mu_i^*(l) + RT \ln \frac{f_i}{f_i^*}$$

Za nizke tlake:

$$\mu_i(l) = \mu_i^*(l) + RT \ln \frac{p_i}{p_i^*} = \mu_i^*(l) + RT \ln x_i = \mu_i(l)$$

za idealne raztopine

Realni sistemi

$$\mu_i(l) = \mu_i^*(l) + RT \ln \frac{f_i p}{f_i^* p^*} = \mu_i^*(l) + RT \ln \frac{f_i}{p^*} = \mu_i^*(l) + RT \ln a_i$$

$RT \ln \frac{p}{p^*}$ pospravimo v μ_i^*

$$a_i = \gamma_i \frac{C_i}{C^*} = \gamma_i \frac{b_i}{b^*} = \gamma_i \frac{x_i}{x^*} \quad \gamma_i = f(T)$$

Termodinamika mešanj

$$\Delta_{meš} X = \underbrace{n_1 \bar{X}_1 + n_2 \bar{X}_2}_{\text{končn. stanje}} - \underbrace{(n_1 X_{1m} + n_2 X_{2m})}_{\text{zač. stanje}}$$

Idealna raztopina

$$\Delta_{meš} G = n_1 \mu_1 + n_2 \mu_2 - n_1 G_{1m} - n_2 G_{2m} = \mu_1(k=g) = \mu_1^* = G_{1m}$$

$$= n_1 \mu_1^* + n_1 RT \ln x_1 + n_2 \mu_2^* + n_2 RT \ln x_2 - n_1 \mu_1^* - n_2 \mu_2^* =$$

$$= n_1 RT \ln x_1 + n_2 RT \ln x_2 \quad / \cdot \frac{n_1 + n_2}{n_1 + n_2}, \quad n_1 + n_2 = n$$

$$\Delta_{meš} G = n \frac{n_1}{n} RT \ln x_1 + n \frac{n_2}{n} RT \ln x_2$$

$$\Delta_{meš} G = n RT (x_1 \ln x_1 + x_2 \ln x_2) < 0 \Rightarrow \text{mešanje je spontano}$$

$$S = - \left(\frac{\partial G}{\partial T} \right)_p$$

$$\rightarrow \Delta_{mcs} S = nR(X_1 \ln X_1 + X_2 \ln X_2) > 0$$

$$H = G + TS$$

$$\Delta_{mcs} H = 0$$

$$\Delta_{mcs} V = n_1 \bar{V}_1 + n_2 \bar{V}_2 - n_1 \bar{V}_{1,m} - n_2 \bar{V}_{2,m} = 0$$

$$\bar{V}_i = \left(\frac{\partial \bar{G}_i}{\partial p} \right)_T = \left(\frac{\partial (\mu_i^* + RT \ln X_i)}{\partial p} \right)_T = \frac{\partial \mu_i^*}{\partial p} + 0 = \bar{V}_{i,m}$$

Neidealna raztopina

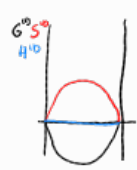
$$\mu_i = \mu_i^* + RT \ln f_i X_i$$

$$\Delta_{mcs} G = n_1 \bar{G}_1 + n_2 \bar{G}_2 - n_1 G_{1,m} - n_2 G_{2,m} = n_1 (\mu_1^* + RT \ln f_1 X_1) + n_2 (\mu_2^* + RT \ln f_2 X_2) - n_1 \mu_1^* - n_2 \mu_2^* =$$

$$= n_1 RT \ln X_1 + n_2 RT \ln X_2 + n_1 RT \ln f_1 + n_2 RT \ln f_2$$

$$= \underbrace{nR(X_1 \ln X_1 + X_2 \ln X_2)}_{\Delta_{mcs} G^0} + \underbrace{nRT(X_1 \ln f_1 + X_2 \ln f_2)}_{\Delta_{mcs} G^{ex}}$$

$$\Delta X^{ex} = \Delta X - \Delta X^{id} \text{ presežna kolizija}$$



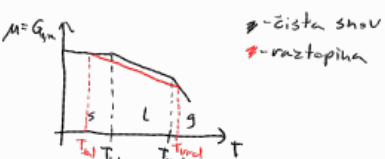
Koligativne lastnosti raztopin

- znižanje parnega tlaka
- znižanje T_{mrt} , zvišanje T_{mel}
- osmotski tlak

$$p_i^*(T) \text{ parni tlak čistega topila}$$

$$p_i(T) = p_i^* X_i = p_i^*(T)(1 - X_2)$$

$$\Delta p_i(T) = p_i^* - p_i(T) = p_i^* X_2$$



$$d\mu_i = -S_m dT + \bar{V}_i dp$$

$$= -\bar{S}_i dT + \bar{V}_i dp$$

$$\mu_i(T) = \mu_i^* + RT \ln X_i$$

$$\frac{\mu_i^*(s)}{\mu_i(l)} = 1 \rightarrow \mu_i^*(s) = \mu_i^*(l) + RT \ln X_i$$

$$RT \ln X_i = \mu_i^*(s) - \mu_i^*(l) = G_{i,n}(s) - G_{i,n}(l) = -\Delta_{fus} G \quad / : T$$

$$R \ln X_i = - \frac{\Delta_{fus} G}{T} \quad / \frac{\partial}{\partial T} \Big|_p$$

$$R \left(\frac{\partial \ln X_i}{\partial T} \right)_p = - \frac{\partial}{\partial T} \left(\frac{\Delta_{fus} G}{T} \right)_p = \frac{\Delta_{fus} H}{T^2}$$

Gibbs-Helmholtz

$$\left(\frac{\partial \ln X_i}{\partial T} \right)_p = \frac{\Delta_{fus} H}{RT^2} \quad / \int dT$$

$$\ln X_1 = \frac{\Delta_{fus} H}{R} \left(-\frac{1}{T} \right) \Big|_{T_2^*}^{T_2} = \frac{\Delta_{fus} H}{R} \left(\frac{1}{T_2^*} - \frac{1}{T_2} \right) \quad \ln X_1 = \ln(1 - X_2) \approx -X_2$$

$$-X_2 = \frac{\Delta_{fus} H}{R} \left(\frac{1}{T_2^*} - \frac{1}{T_2} \right) \rightarrow X_2 = \frac{\Delta_{fus} H}{R} \frac{\Delta T_2}{T_2^* T_2} \approx \frac{\Delta_{fus} H}{R} \frac{\Delta T_2}{T_2^{*2}}$$

$$\Delta T_2 = T_2^* - T_2 \quad X_2 = b M_1$$

$$bM_1 = \frac{\Delta_{\text{fus}} H \cdot \Delta T_2}{R(T_2^*)^2}$$

$$\rightarrow \Delta T_2 = \frac{M_1 R T_2^{*2}}{\Delta_{\text{fus}} H} b = k_K b$$

↳ krioskopska konst.

$$\Delta T_2 = k_K b$$

$$K_K = \frac{M_1 R T_2^{*2}}{\Delta_{\text{fus}} H}$$

urešnice:

$$\mu_1^*(l) + RT \ln x_1 = \mu_1^*(g)$$

$$RT \ln x_1 = \mu_1^*(g) - \mu_1^*(l) = G_{1,g}(g) - G_{1,g}(l) = \Delta_{\text{izp}} G \quad \frac{1}{T}$$

$$RT \ln x_1 = \frac{\Delta_{\text{izp}} G}{T} \left/ \frac{\partial}{\partial T} \right|_p$$

$$R \frac{\partial \ln x_1}{\partial T} = - \frac{\Delta_{\text{izp}} H}{T^2} \quad \bigg/ \int dT$$

$$\ln x_1 \approx -x_2 = - \frac{\Delta_{\text{izp}} H}{R} \frac{\Delta T_2}{(T_2^*)^2}$$

$$\Delta T_2 = K_e b$$

↳ ebulioskopska konst.

$$K_e = \frac{R M_1 (T_2^*)^2}{\Delta_{\text{izp}} H}$$

Osnova

T = konst.	
topilo	raztopina
p	p + Δp

(topilence ne sme prebijati skozi mejo)

$$\mu_1(p, T) \quad \mu_1(p + \pi, T, x_1)$$

Δp = π osmotski tlak

$$\mu_1^*(p) = \mu_1(p + \pi, x_1)$$

$$\mu_1(p + \pi, x_1) = \mu_1(p, x_1) + \int_p^{p+\pi} \left(\frac{\partial \mu_1}{\partial p} \right) dp$$

$$\mu_1^*(l) + RT \ln x_1$$

• razredčena vzet.: $\bar{V}_1 \approx \bar{V}_{1,m}$
• nestisljivost: $\bar{V}_{1,m} \neq f(p)$

$$\rightarrow \mu_1^*(l)(p) = \mu_1^*(l)(p) + RT \ln x_1 + \bar{V}_{1,m} \pi$$

$$RT \ln x_1 = - \bar{V}_{1,m} \pi$$

$$RT x_2 \approx \bar{V}_{1,m} \pi$$

$$\pi = \frac{RT x_2}{\bar{V}_{1,m}} = \frac{RT n_2}{n_1 \bar{V}_{1,m}} = \frac{RT m_2}{V_1} \approx \frac{RT n_2}{V} = c RT$$

$$\boxed{\pi = c RT}$$

van't Hoff

• če topilence razpade v topilo na i delcev:

vse enačbe se na desni pomnožijo z i

$$\Delta T_2 = i K_K b$$

$$\Delta T_2 = i K_e b$$

$$\Delta p = i p_1^* x_2$$

$$\pi = i c RT$$

Primer: izračun koligativnih lastnosti za fiziološko razt.

(0,15 M NaCl)

$$T = 25^\circ \text{C}, \quad p_1^* = 3170 \text{ Pa} \quad (\text{H}_2\text{O})$$

$$b = \frac{n_2}{n_1 M_1} = \frac{c}{M_1} \quad 0,15 \text{ mol} \cdot \frac{\text{kg}}{\text{kg}} \quad i = 2$$

$$\Delta p = i p_1^* x_2 = i p_1^* b M_1 = 2 \cdot 3170 \text{ Pa} \cdot 0,15 \cdot \frac{\text{mol}}{\text{kg}} \cdot 0,018 \frac{\text{kg}}{\text{mol}} = 17 \text{ Pa}$$

$$K_K = \frac{R M_1 (T_2^*)^2}{\Delta_{\text{fus}} H} = 1,86 \text{ K} \cdot \frac{\text{kg}}{\text{mol}} \rightarrow \Delta T_{\text{kel}} = K_K \cdot b = 0,56 \text{ K}$$

$$K_e = \frac{R M_1 (T_2^*)^2}{\Delta_{\text{izp}} H} = 0,513 \text{ K} \cdot \frac{\text{kg}}{\text{mol}} \rightarrow \Delta T_{\text{mel}} = i K_e b = 0,15 \text{ K}$$

$$\pi = i c RT = 7,44 \text{ bar}$$

Raztopine plinov v vodi

$$k_H(\text{O}_2) = 3,3 \cdot 10^7 \text{ torr} \quad (1 \text{ torr} = 133,3 \text{ Pa})$$

$$k_H(\text{N}_2) = 6,51 \cdot 10^7 \text{ torr}$$

$$k_H(\text{CO}_2) = 1,25 \cdot 10^6 \text{ torr}$$

voda je v stiku z zrakom, p = 1 bar

↳ 20% O₂, 80% N₂, 400 ppm CO₂

$$p(\text{O}_2) = k_H(\text{O}_2) x_2(\text{O}_2)$$

$$\downarrow$$

$$0,2 \cdot p = 20 \text{ kPa}$$

$$x_2(\text{O}_2) = 4,55 \cdot 10^{-6}$$

$$b = x_2 / M(\text{O}_2) = 4,42 \cdot 10^{-4} \frac{\text{mol}}{\text{kg}}$$

↳ 32 g/mol

$$x_2, b, c = ?$$